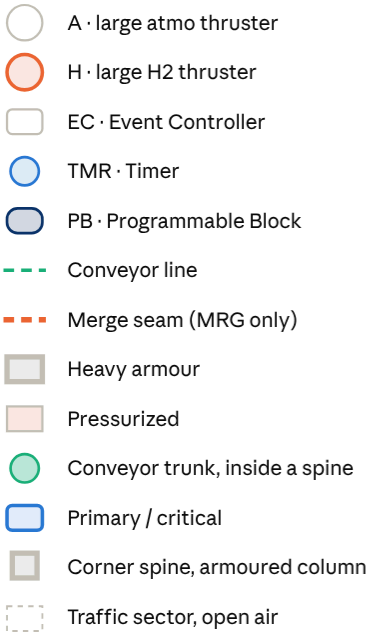
Floating Rig Build Guide

Oct 15, 2026 · @Blae

Key

Key: every symbol and abbreviation used in this guide

Symbols



Structure

HUB hex core hub
H1-H6 hub decks, top down
P-E/NE/NW/W/SW/SE pylons
T-N/ENE/ESE/S/WSW/WNW sectors
B large block, 2.5 m
kt MN MW kilotonne, MN, MW
MNR ICE RLY miner, ice, relay
REP FAB repair, fabricator
INT PKT interceptor, picket

Blocks

R large reactor
BAT battery
H2E hydrogen engine
GEN O2/H2 generator
ASM assembler
CON MRG connector, merge
SRT PRJ sorter, projector
WLD SNS welder, sensor
MC RC main cockpit, remote
AR BC action relay, broadcast

Logic

∧ ∨ AND gate on, off
→ action
SP% PO% stored, output pwr
CF% GT% cargo, gas tank fill
T% BI% thrust, integrity
CR CC conn. ready, connected
DO BR door open, block gone
U10 U100 PB tick rate

Power and modes

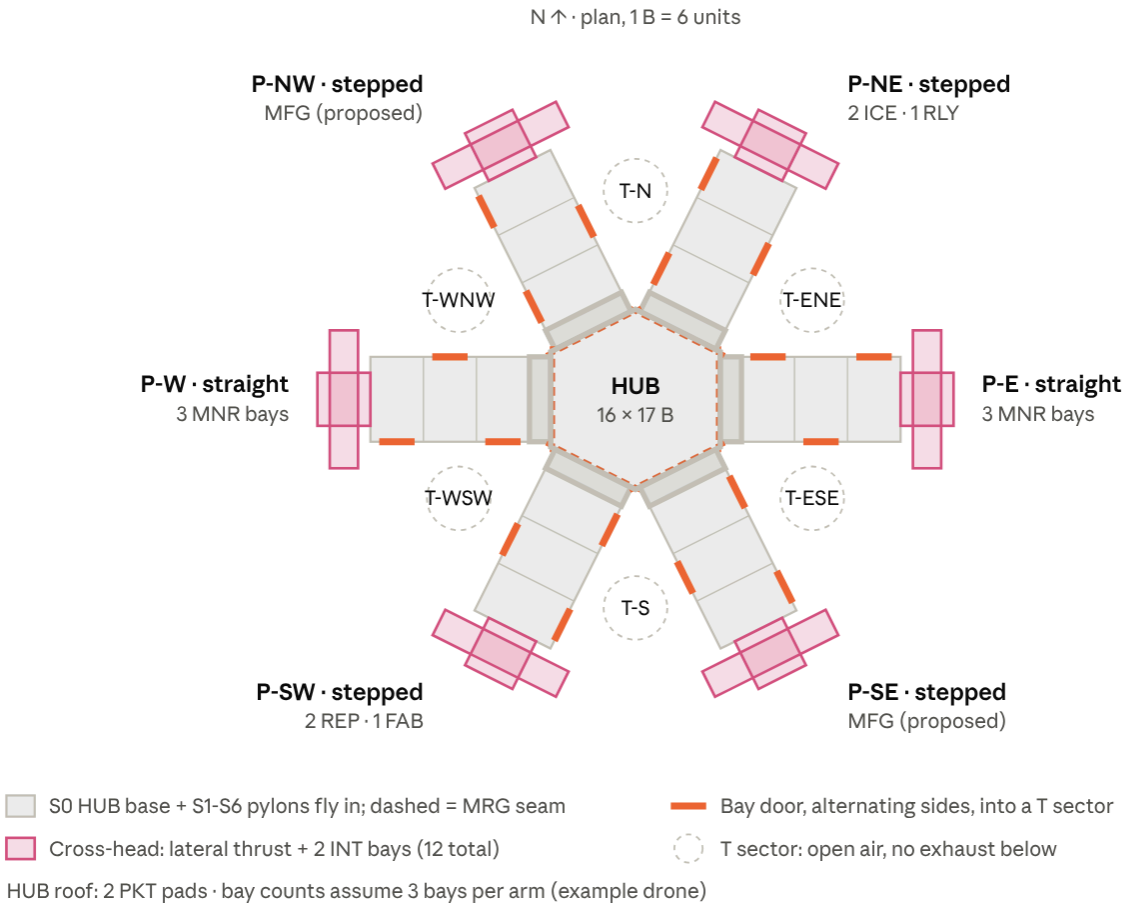
WB RB working, reserve bank
RCH DIS AUTO battery mode
SPK tank stockpile
NRM CAU SHD EMG modes
CBT REP REC modes
† verify in creative

key sheet · symbols and abbreviations

- Naming: `[ID] Fn n`, e.g. `[P-NE] A 3`.
- Groups per ID: `Atmo Lift` · `H2 Lift` · `Reserve Tanks` · `Lateral` · `Ops` · `Shed` · `Repair`.
- Rig-wide groups: `All H2 Lift` · `All Reserve Tanks` · `All Shed` · `[HGR] Doors` · `[HGR] Connectors`.
- Figures ≈, example rig 20 kt @ 1 g. All DLC available.

Rig overview

Six pylon arms hold the hangars; the six gaps between them stay open for traffic



rig plan · hex HUB, 6 P, 6 T sectors

10 sections, each designed and built on its own: S0 HUB base (H4–H6) flies first → S1–S6 pylons dock in opposite pairs (E + W, NE + SW, NW + SE) → S7–S9 decks H3, H2, H1 stack on the base · every seam MRG + L CON · T sectors stay empty.

Rules

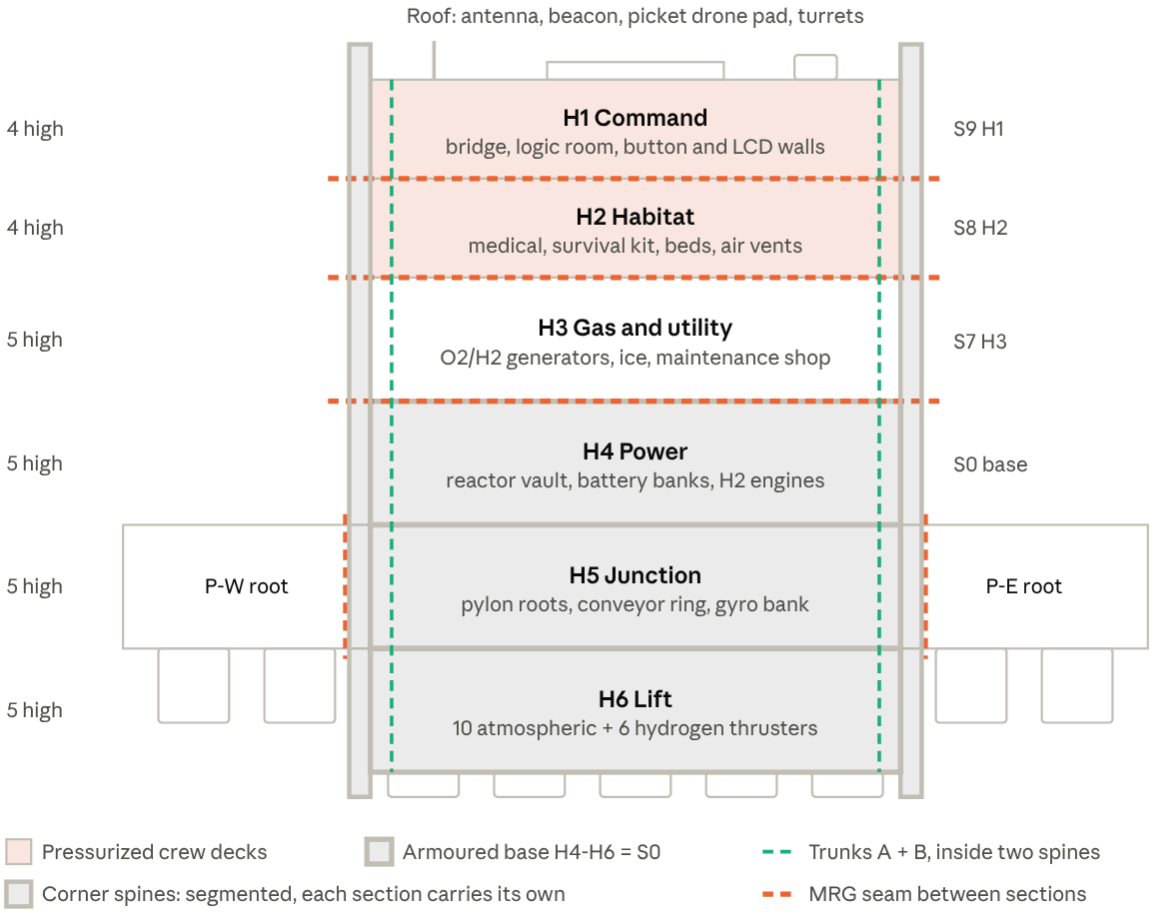
- **R1** Practical only: each block carries load, moves material, makes power or protects.
- **R2** Mass = cost. Light armour default; heavy on R vault, MC, SPK tanks, P roots, corner spines.
- **R3** Pressurize H1–H2 only.
- **R4** A carries hover @ ≈58% T% (NRM); HUB ≈19% of A lift, each P ≈13%.
- **R5** Blackout: H alone hovers @ ≈91%; H OFF until EMG.
- **R6** Any 2 P lost → hovers on A (≈80%); any 3 lost → A + H.
- **R7** A + H on every lift surface.

- **R8** Every flown section (HUB base, P) carries a seed kit: own lift, RC, gyros, BAT, antenna. Stacked decks carry none.
- **R9** Every section seam = MRG + L CON; no welds across; any section can be released.
- **R10** 2 conveyor paths to every critical system.
- **R11** Exhaust clear of decks, bays and lanes; T sectors = exhaust-free shafts.
- **R12** Native blocks = safety; PB = decisions.
- **R13** Test every section in creative before survival.
- **R14** One grid: diagonal P run stepped 1:2, clad in 1×2 slopes; nothing off-grid.

Core hub (HUB)

Hex 16 × 17 B (flat E–W, vertex N–S, 1×2 slope faces) · 28 B (70 m) · 6 corner spines H6 → roof · H4–H6 armoured base = S0 (flown) · H3, H2, H1 = stacked sections S7–S9.

The hub stacks six decks between six armoured corner spines

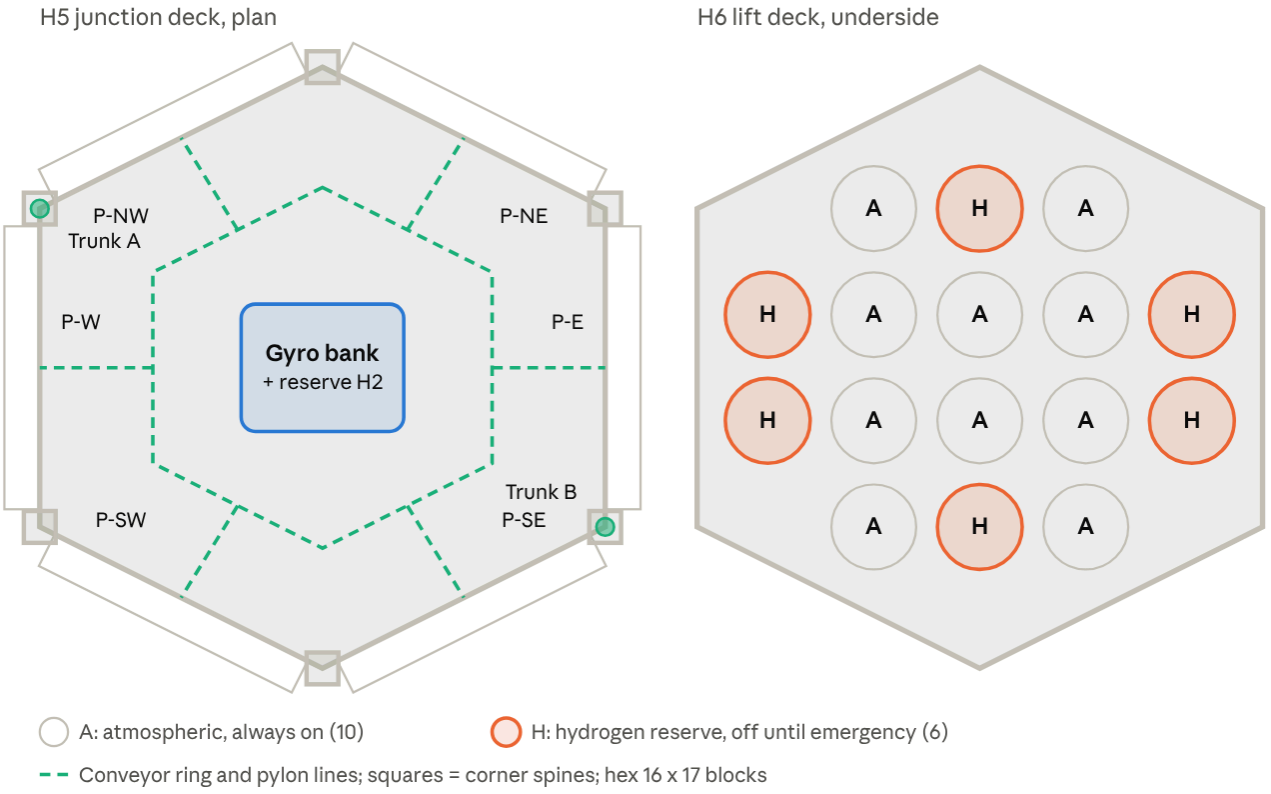


HUB elevation · H1-H6, 28 B, not to scale

- P sockets @ H5, one per hex face: flat MRG plate + 2 L CON; diagonal sockets face N or S → P and HUB thrusters ≈ one plane.
- R, tanks, gyros low and central (base).

- Spines: heavy armour, H6 → roof, split at every seam; each section carries its own segment. Trunk A in WNW spine, B in ESE spine; joined across seams by L CON; either alone feeds every deck. Other 4: ladders, spare trunks.

H5 ties six pylons into one ring; H6 carries a fifth of the atmospheric lift



H5 junction plan · 6 P roots · H6 lift slots (10 A + 6 H)

- HUB lift: 10 A (≈65 MN) + 6 H (≈43 MN†) on 16 slots, 3 × 3 B face; the hex fits 16, not 25.

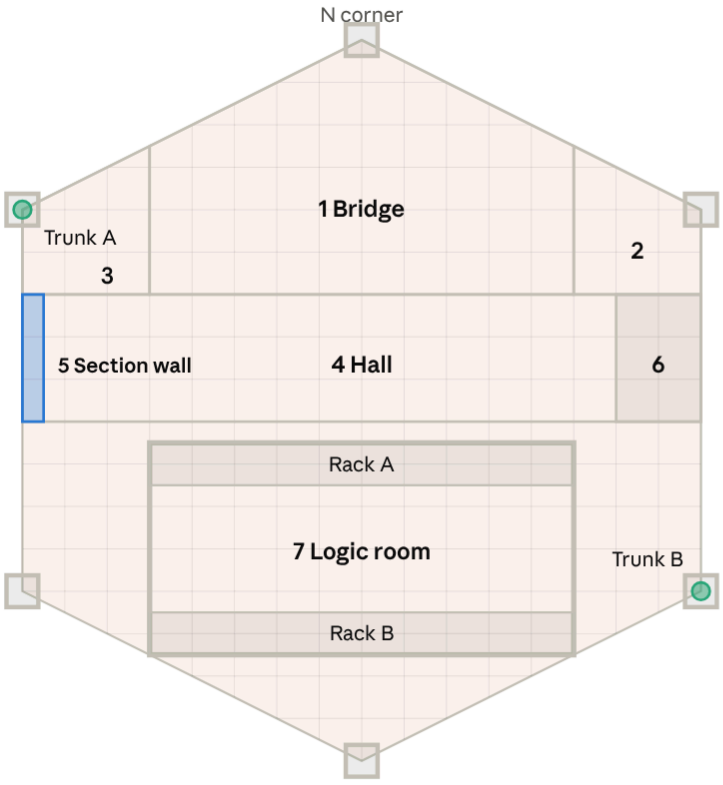
Build

1. Stand or scaffold: H6 clears ground.
2. H6 → H5 (gyros, ring) → H4 (1 R online, RC + antenna on H5 = S0 seed kit).
3. Hover test, base only.
4. Dock P × 6 in opposite pairs: E + W → NE + SW → NW + SE (Pylon).
5. Stack H3 → H2 → H1: each by PRJ from its own blueprint, MRG seam below.
6. Commission → climb to station altitude.

H1 Command

Pressurized · 4 B · MC, comms, panels, logic racks.

H1 puts every control the crew touches on one pressurized deck



1 Bridge
Main Cockpit, LCD wall, mode consoles each side

2 Comms
Broadcast Controller, Action Relay, antenna feed

3 Service corner
Trunk A riser, tool lockers, spare parts

4 Hall
air vents; joins every room

5 Section wall
one four-button panel per section

6 Access shaft
ladder to H2, hatch to the roof airlock

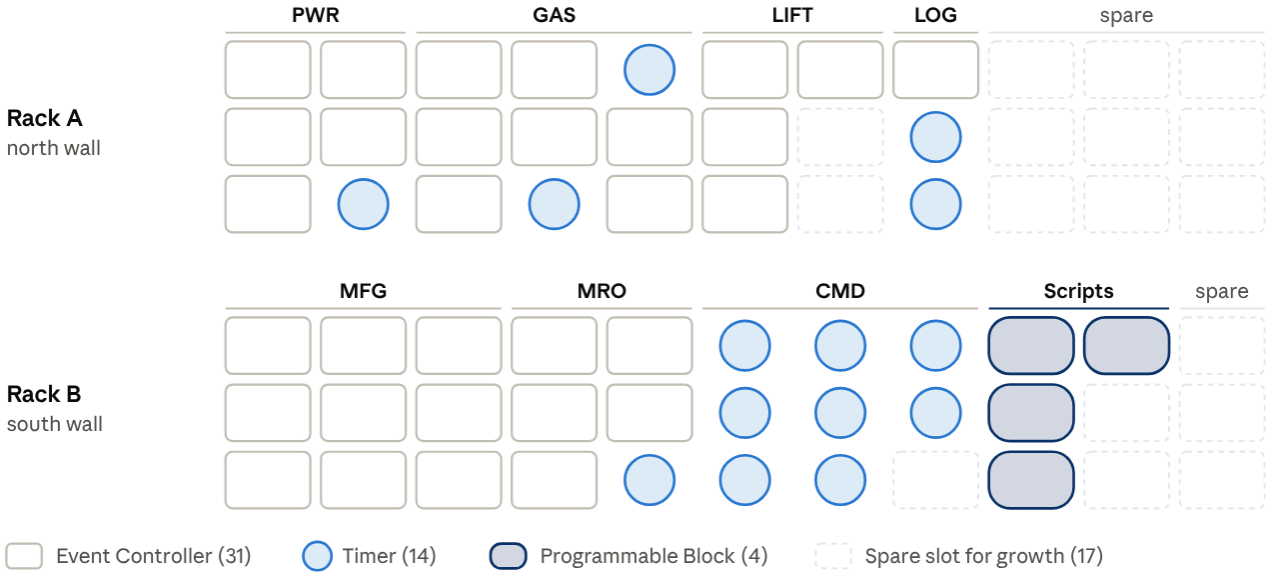
7 Logic room
heavy armour box; Racks A and B hold all ECs

Grid: 1 square = 1 large block (2.5 m); hex deck 16 x 17, 4 blocks high

Corner squares: armoured spines; circles: trunks A and B, continuous through every deck

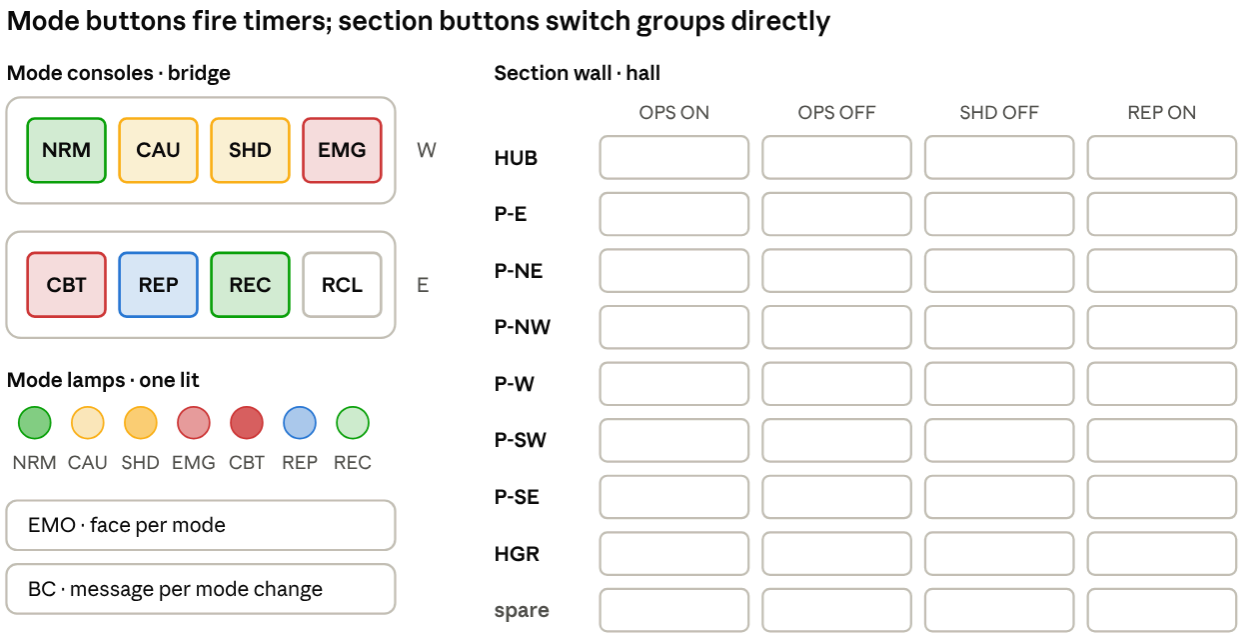
H1 plan · 7 zones, hex 16 × 17 B

Every Event Controller and Timer has a fixed rack slot, grouped by subsystem



H1 rack map · 31 EC, 14 TMR, 4 PB, 17 spare

- Logic room: heavy-armour box. Slot order = ladder rung order.



H1 panel layout · 2 mode consoles, 9-row section wall

- Status: LCD apps (Energy & H2, Horizon, Gravity, Weather†) · mode lamps · EMO · BC.

Block	n
MC or Bridge Helm	1
LCD	6–8
Light (mode lamp)	7
Emotion Controller	1
4-button panel	11
BC · AR · antenna · beacon	1 each
EC · TMR · PB	31 · 14 · 4
RC (backup)	1
Air vent	2–3

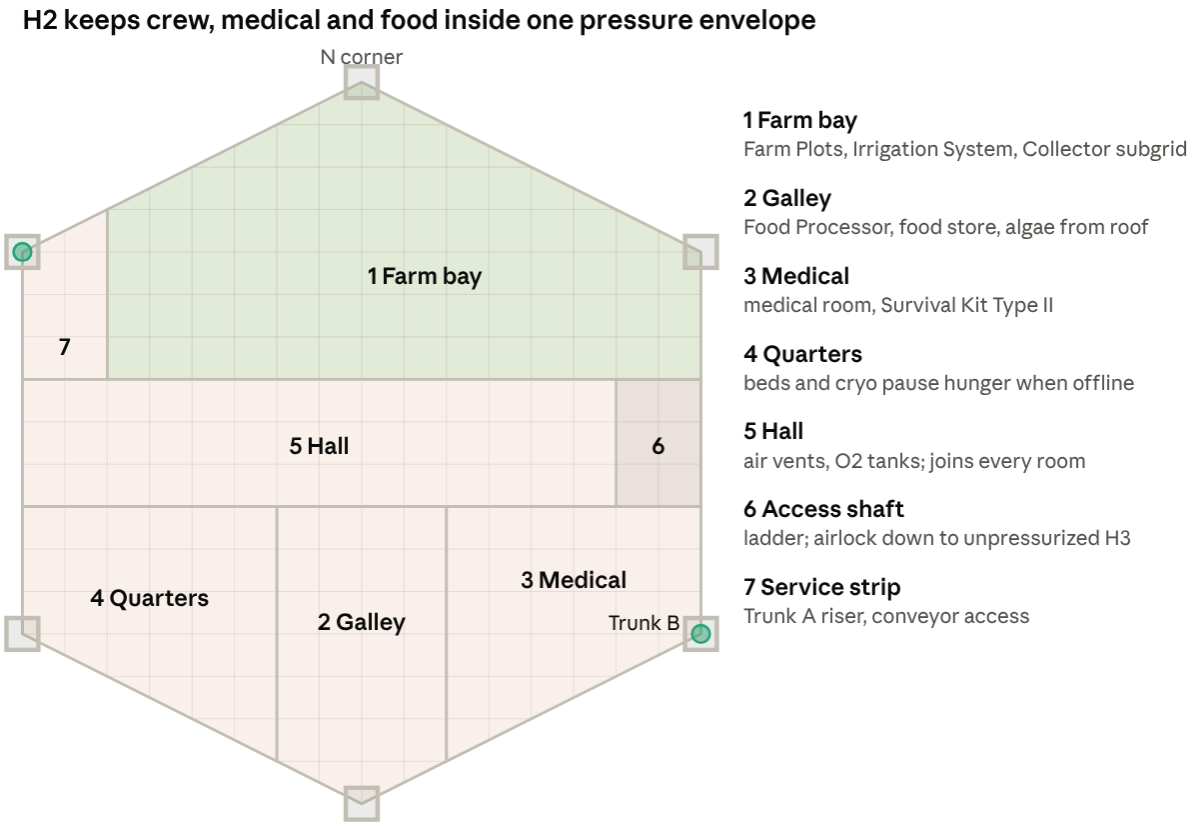
Build

1. PRJ above H2; airtight; armour logic room.
2. Racks in slot order; name before wiring.
3. CMD TMRs → ECs in rung order.
4. MC, LCDs, lamps, EMO.

5. Panels.
6. Comms.
7. Pressurize; test every mode button.

H2 Habitat

Pressurized · 4 B.



H2 plan · 7 zones, hex 16 × 17 B

food chain · algae and crops to meals

- ☢ Damaged R: r ≈ 60 m through structures → crew > 60 m until repaired.

Block	n
Medical room · Survival Kit II	1 · 1
Bed / cryo	1 per crew + spare
Food Processor · cargo S (food)	1 · 1

Block	n
Farm Plot	n
Irrigation System	1
Collector (locked hinge)	1–2
O2 tank · air vent	2 · 3–4
Algae Farm (H1 roof)	n

Build

1. PRJ above H3; airtight; airlock ↓ H3.
2. Vents + O2; pressurize.
3. Medical + kit (respawn).
4. Quarters.
5. Galley + 2 EC.
6. Farm bay.
7. Algae on H1 roof.

H3 Gas & utility

Unpressurized · 5 B.

H3 plan · 6 zones, hex 16 × 17 B

gas flow · ice → O2 + H2 → consumers

- Shop: ASM 1 + refinery 1 → core self-sufficient before MFG arms.
- Core PRJ: HUB + P blueprint, Keep Projection, OFF until REP. Rebuilds within WLD reach only → WLD arrays on H6 (A/H), H4 (R), H5 (gyros).
- H3 lost → no new gas; H5 + P reserves hold on SPK until EMG.

Block	n
GEN	4+ (until H2 work ≥ 70% in NRM)
Cargo L (ice)	2
H2 tank L (working) · O2 tank	4 · 2

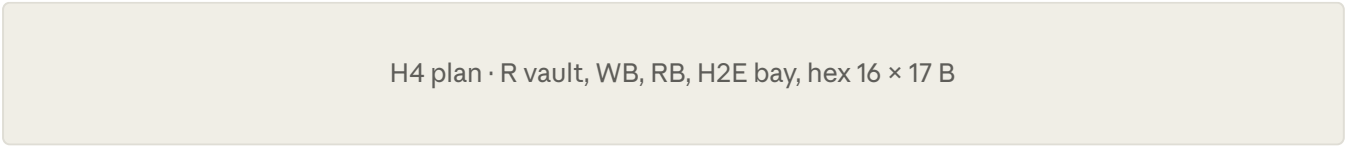
Block	n
SRT	≈ 8
ASM · refinery	1 · 1
Cargo M (stock)	3–4
PRJ	1

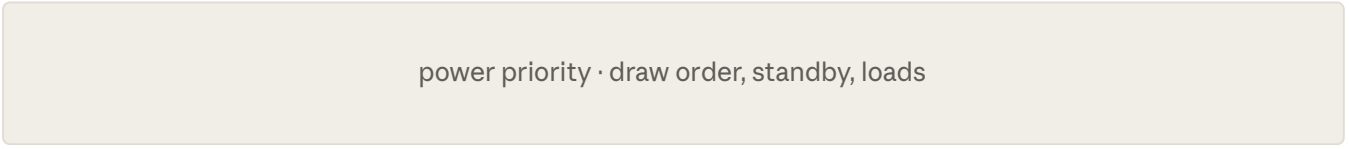
Build

1. PRJ above H4.
2. Manifold: junctions to trunks A + B, SRT bank.
3. Ice + GEN; verify fill.
4. Tank farm (armoured); fill H5 reserve → SPK.
5. Shop + quota ECs.
6. PRJ: load, OFF.
7. Airlock ↑ H2.

H4 Power

Armoured base · 5 B.





Item	Value
A power	16.8 MW / 6.48 MN → ≈ 2.6 MW/MN
Hover (196 MN)	≈ 510 MW
+1 kt mass	≈ +25 MW
Full A (52 A)	≈ 874 MW
R × 3	900 MW · N−1 = 600 MW ⩾ hover
U @ hover	≈ 510 kg/h (1 kg = 1 MWh)
BAT	12 MW · 3 MWh

Item	Value
H2E	5 MW · 500 L/s

- Lift on BAT or H2E: no (> 40 BAT, ≈ 100 H2E). Blackout → H lift (R5).
- WB: RCH ⩽ 20%, AUTO ⩾ 95%. H2E OFF until R fail. RB: RCH; DIS on R fail.
- ☢ r ≈ 60 m → vault WLD ring in REP.

Block	n
R (3 × 3 × 3 B, 74 t)	3
WLD (vault ring)	≈ 8
Cargo + SRT (U)	1 + 1
BAT WB · RB	8 · 6
H2E	4

Build

1. PRJ above H5.
2. Vault shell → R × 3 + U links.
3. Load U; 1 R online.
4. WB AUTO · RB RCH.
5. H2E OFF.
6. WLD ring OFF.
7. Wire PWR rungs + F3.

Pylon template (P × 6)

≈ 13% of A lift each · flown section: built on a ground pad, flies up on its own lift, docks at an H5 socket · drone hangars in the arm · defence bays in the cross-head · 3 hand builds (P-E, P-NE, P-NW), 3 by PRJ rotated 180°; PRJ cannot mirror.

P plan + section · bays, keel, cross-head

P docking · 4 moves at the H5 socket

Item	Spec
Section	W = bay depth = drone L + 4 B, max 9 B (hex face: wider arms collide at the root) · H = 1 B deck + bay H + 2 B keel (example 8 × 7 B)
L	2 B collar + bays × bay W + 5 B cross-head (example 3 × 5 + 7 = 22 B, 55 m)
Bays	standard bay, full arm width; doors alternate sides → both flanking T sectors; 3–4 per arm
Keel	under the bays: 2 conveyor lines, H2 tanks (heavy, SPK)
A	7 · ≈ 45 MN · always ON
H	4 · ≈ 29 MN† · OFF until EMG
Cross-head	plus-shaped tip: A out + both sides (lateral), H out; 2 INT bays, roof hatches; turrets top + bottom; CON pad
Root	MRG plate + 2 L CON on the H5 socket; heavy armour collar
Stepped P	NE, NW, SW, SE: run 1:2 on the hub grid, 1×2 slope cladding, bays stay square and stagger
Seed kit	RC 1 · gyro 2 · BAT bank† · antenna; lifts the arm alone at T% < 80†

Build

1. Root collar, heavy armour; MRG plate + 2 L CON facing the socket.
2. Keel to L: 2 conveyor lines → HUB ring, tanks, SPK.
3. Bay row over the keel: floors, walls, CONs, doors on alternating side walls.
4. A, then H (H disabled).
5. Cross-head: lateral A + H, INT bays, turrets, CON pad.
6. X-link conveyors at tip.
7. Seed kit.
8. Name + groups.
9. Blueprint → PRJ: E → W, NE → SW, NW → SE. Fly up, dock (sequence above).

Tests

- [ ] 1 P A OFF → CAU, altitude holds.
- [ ] 2 opposite P OFF → hovers on A.
- [ ] 3 adjacent P OFF → EMG, A + H hold.
- [ ] 1 conveyor line cut → H2 still flows.
- [ ] Drone out of each bay door → T sector, no exhaust contact.
- [ ] Release MRG (creative) → P hovers on its seed kit†.

Traffic sectors (T × 6)

T sector · plan + section, exhaust-free shaft

- T-N, T-ENE, T-ESE, T-S, T-WSW, T-WNW: the open air between pylons, centred on the HUB corners. No structure, ever.
- Lift exhaust falls only under the arms and the HUB → each T is a clear vertical shaft to the surface and to orbit.
- Each T serves the bay doors of its 2 flanking arms: 1 descent lane, 1 climb lane; Dispatcher queues per T.
- T-N and T-S are narrower (≈53° vs ≈63°): a 1×2-slope hex is not quite regular.

Hangars (HGR)

bay board · 6 arms, 12 INT, 2 PKT (proposal)

- Work drones: bays in the arms, doors on side walls into the flanking T. Defence: INT bays in each cross-head, roof hatches. Pickets: 2 pads on the HUB roof.
- Nothing opens ↓ (exhaust). Before bays are fitted: drones on tip CON pads.

standard bay · plan, not to scale

- Bay = drone bbox + 1 B all round + 2 B ahead of CON. WLD + SRTs always ON.
- Local logic: rung HG1 (1 EC/HGR) · SNS/bay (doorway → door OPEN) · watchdog → `[HGR] Doors` OPEN.
- Launch/recovery: IGC sequence, Scripts.

Control modes

mode ladder · T% A bands, not to scale

Mode TMRs NRM · CAU · SHD · EMG · CBT · REP · REC: set states only; REC via TMR delay. F1–F3 (H5) never disabled.

Automation & wiring

automation stack · sense, decide, act, report

subsystem interconnection · flows, power, command

EC wiring ladder · 33 rungs, 8 groups

- **W1** 1 event per EC; pair ECs for ↑/↓ thresholds. **W2** Set actions only (ECs can re-trigger). **W3** > 1 action → TMR, Trigger Now. **W4** ∧ all, ∨ any. **W5** EC sees own + merged/connected grids; AR for the rest. **W6** `[CODE] EC sig` · `[CODE] TMR resp`.

quota cell · rungs M1 + M2

- CF% reads the whole container → 1 product per container.

drone cycle · rungs D1-D4, native fallback

combat chain · 3 sources, TMR CBT, 5 responses

- Decoys on P tips.

Scripts (PB)

control layers · 4 rig PBs over the native layer

PB	Role	Rate	Acts via
`[CMD] PB Rig OS`	lift margin · modes · anchor hold · heartbeat	U10	mode TMRs
`[MFG] PB Inventory`	quotas · sorting · ASM queues	U100	ASM, SRT
`[LOG] PB Dispatcher`	jobs · bays · launch/land queue	msg + U100	IGC, doors, CON
`[CMD] PB Displays`	dashboards	U100	LCD

- Drone PBs: miner (PAM-like) · ice (miner variant) · interceptor (script or AI Offensive) · picket (none: AI Defensive + AR) · repair (later). All keep D1–D4 + AR RECALL.

IGC sequence · launch, recovery, broadcast

- **S1** Native safety never depends on a PB. **S2** Watchdog TMR 30 s; Rig OS Stop → Start each run; 0 → BC + fallback ECs ON. **S3** Fallback ECs OFF while PBs healthy. **S4** Act via TMRs and groups. **S5** Config in Custom Data. **S6** Cache blocks; U100 default. **S7** 1 rack slot per PB.
- Off-the-shelf: Isy's Inventory Manager · MMaster's Automatic LCDs 2.

Commissioning (per section)

- [ ] MRG engaged; no welds across seam.
- [ ] Conveyor round trip, 1 item.
- [ ] Names `[ID]`; groups complete.
- [ ] Panel row assigned.
- [ ] Section RC OFF; MC in control.
- [ ] T% A in NRM band.
- [ ] EMG test: sheds, H ON.
- [ ] REC via TMR delay.

- [ ] Release test (creative)†.

Open

- [ ] Planet (assumed Earthlike, 1 g).
- [ ] Survival or creative.
- [ ] Altitude band.
- [ ] Target mass (example 20 kt; hangar arms add mass).
- [ ] Drone sizes → bay size → arm section and P length (drone L ⩽ 5 B fits the 9 B arm limit).
- [ ] MFG home: P-NW + P-SE arms (proposed) or HUB.
- [ ] Deck sections: stacked by PRJ on the base (current) or delivered by tug; swap procedure.
- [ ] Bay allocation (board = proposal).
- [ ] H5, H6 deck chapters.
- [ ] Mass reduction pass.

Verify†

- [ ] Thrust acts through CoM (1 P OFF, no roll).
- [ ] Severed P keeps dampeners.
- [ ] H2 drain time at full H.
- [ ] Dampener drift vs drone routes.
- [ ] Consumers pull through SRT.
- [ ] Large H thrust value.

Log

Date	Change
2026-10-16	Every module a section: S0 HUB base, S1–S6 flown pylons docking at H5 MRG sockets, S7–S9 stacked decks
2026-10-16	Hex core + corner spines; 6 P with hangar arms and INT cross-heads; T sectors replace Q; lift 52 A + 30 H, hover now in NRM (was ≈65%, CAU band)
2026-10-15	Refactor: spec sheets, EC ladder, key, new drawings
2026-10-15	Guide created

Sources

- Event Controller · Large Atmospheric Thruster · Large Reactor · Battery · Hydrogen Engine (Space Engineers Wiki)
- Apex Survival · Update 1.207 notes (Keen Software House)